

Conformalized approach for classification intervals

How to get intervals for binary classification tasks?



While there are many approaches to quantifying uncertainty with conformal prediction, for binary classification tasks (e.g., credit scoring, fraud detection) practitioners are typically left with a confidence set of **[True, False]**

In real life this leads to some comical situations such as the one depicted below, [an example](#) I borrow from a leading expert on the subject, Valery Manokhin



We don't typically care about prediction sets, we care about some downstream decision ([quote](#) by Anastasios Angelopoulos)

For regression problems, in [his paper](#) Ryan Tibshirani offered a studentized residual as a residual conformity score where the standard deviation is estimated separately

A classification alternative to this can be the Pearson residual:

$$r_i = \frac{y_i - \hat{p}_i}{\sqrt{\hat{p}_i(1 - \hat{p}_i)}}$$

where y_i is the observed outcome and p_i is the predicted probability from a classifier

A chosen confidence (e.g. 95%) quantile of absolute Pearson residuals on the calibration set is then used to construct the conformalized classification intervals:

$$q_\alpha = \text{Quantile}_{1-\alpha}(|r_{\text{calibration}}|)$$

$$\text{Lower Bound (LB)} = \max\left(0, \hat{p}_i - q_\alpha \times \sqrt{\hat{p}_i(1 - \hat{p}_i)}\right)$$

$$\text{Upper Bound (UB)} = \min\left(1, \hat{p}_i + q_\alpha \times \sqrt{\hat{p}_i(1 - \hat{p}_i)}\right)$$

Conformalized approach for classification intervals

Application of the approach to real-world data



In this example, we use a real-world dataset (breast cancer) to visualize the 95% confidence intervals for four well-established algorithms for binary classification:

- Logistic Regression
- Naïve Bayes
- SVM
- Random Forest

The plots illustrate how the intervals reflect the actual coverage of points above or below the predicted probabilities, allowing us to evaluate and compare models that produce probability-like scores

While this experimental approach is not a classical case of conformal prediction (CP), it shares a similar objective of quantifying uncertainty using established methodologies from both machine learning (CP) and traditional statistics (discrepancy measures)

[Notebook](#)

